Mapping the Margins: Building a Cross-National Observatory for Alternative News Media Research

Abstract

This chapter introduces a digital, comparative platform designed to collect, analyse, and visualize alternative news media (ANM) content across national contexts. The concept is called an *Alternative Media Observatory* and is grounded in the notion of "distant data". The observatory offers a novel methodological and epistemological framework for examining fringe information environments at scale, while remaining sensitive to their discursive and ideological complexity. The observatory supports both scholarly and journalistic inquiry, as well as student engagement, by providing structured access to ANM corpora through ethically collected, computationally processed, and interactively visualized data. Drawing on a cross-national pilot study of ANM in Denmark, Finland, Norway and Sweden, the chapter outlines the full lifecycle of observatory construction – from outlet selection and web scraping protocols to multilingual content extraction, topic modeling, and interface design. By bridging qualitative interpretation with scalable, reproducible methods, the Observatory enables researchers not only to explore how alternative media articulate political, cultural, and epistemic positions but also facilitates hypothesis generation and systematic comparative inquiries into cross-national ANM dynamics. The chapter concludes by reflecting on the ethical considerations of scraping politically sensitive content and argues for transparent, reflexive infrastructures that advance comparative ANM research.

Keywords: data analysis; data visualization; media observatory; comparative research; alternative news media; network analysis; topic modelling

Introduction

Alternative news media (ANM) research is a rapidly evolving interdisciplinary field at the intersection of political communication, media studies, and journalism. Its epistemic goals and methodological approaches remain open to development, shaped by shifting media ecosystems and ongoing debate about what constitutes "alternative" or "mainstream". Much of the foundational work in the field has relied on qualitative case studies, which have provided crucial insights into the stylistic, argumentative, and ideological dimensions of ANM content (see e.g. Figenschou & Ihlebæk, 2019; Frischlich et al., 2020; Hameleers, 2025; Mayerhöffer, 2021; Nygaard, 2019; Pedersen et al., 2024). This body of work has been crucial in establishing

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ANM research as a field by revealing how alternative media outlets shape media and political discourse by challenging mainstream media logics.

Yet as the field matures, the limits of case-based, single-country studies become more apparent – particularly in addressing the transnational, networked nature of digital ANM ecosystems. To advance ANM research within the broader landscape of communication and media studies, there is a need for comparative approaches that draw on computational and quantitative methods. Country-comparative approaches are rare in existing works – with few notable exceptions (Heft et al., 2021; Henriksen et al. 2024; Kristensen et al. 2023; Nygaard, 2019; Staender et al. 2024) – and can reveal recurring patterns and important divergences across national media systems, political cultures, and regulatory environments (see also chapter 16 in this volume). For instance, comparative designs can help determine whether key features – such as partisan or conspiratorial framing – are best explained by global information dynamics or by local historical trajectories.

Commented [3]: dont anonymize references. and true, there arent many studies, but certainly more than these...Consider whether you limit yourself to comparative studies drawing on quantitative and computational methods - then you only need to include a few more, e.g. Heft et al., 2021 or Staender et al., 2024

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Moreover, the field lacks unified frameworks for systematically mapping actors, narratives, and issues – such as migration or climate policy – across multiple ANM contexts. To address this gap, we need infrastructures that support scalable, cross-border analyses while maintaining interpretive depth. One such proposal is the development of an observatory model: a digital platform for collecting, analysing, and visualising ANM content that supports both distant, data-driven comparisons and close, qualitative interpretation. Inspired by the "distant reading" approach (Moretti, 2013), the observatory model adopts a "distant data" perspective: it favours systemic over textual analysis, treats abstraction as a productive constraint, and foregrounds the full archival field of alternative media – enabling researchers to detect emergent patterns and comparative structures that case studies of prominent ANM cannot yield.

In response to this need, this chapter introduces the *Alternative Media Observatory* – an open, interactive platform for researchers, students, and journalists to explore and engage with ANM content across national contexts. By integrating qualitative, quantitative, and computationally oriented methods (including NLP, topic modelling, and network analysis), the observatory provides a flexible, scalable infrastructure that offers a synchronized cross-national overview and enables topic- and actor-centred mapping through systematic tagging, computational text analysis, and dynamic visualizations. It supports hypothesis generation by coupling automated insights with sampling of texts for close qualitative reading, thereby seeking to advance ANM research for the investigation of alternative opinion dynamics and information dissemination. Beyond mapping discourse, the observatory's quali-quant design also invites broader investigations: tracing how issues travel across borders, following the diffusion of frames over time, comparing agenda-setting between ANM and mainstream outlets, or linking content dynamics to audience interaction. Such uses position the observatory not only as a tool for analysing texts, but as an entry point into the study of transnational flows, networked publics, and shifting political cultures.

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Designed to bridge the deep insights of qualitative inquiry with the scale and reproducibility of computational approaches, the observatory remains accessible even to users with limited programming experience. To illustrate this in practice, the chapter presents a concrete implementation case, based on ANM from four Northern European countries (Denmark, Finland, Norway, and Sweden). This also includes a discussion of the ethical dimensions that arise when collecting, analysing, and publishing politically sensitive material produced by actors who may oppose scrutiny, resist categorization, or strategically obscure their public-facing roles. The chapter emphasizes the importance of responsible data collection, transparent

methodological choices, and reflexive interpretation – especially when dealing with content that may simultaneously contribute to valid societal critique and delegitimization of democratic institutions.

Conceptual Considerations: "Distant" Data?

Alternative news media can be defined broadly as media outlets that explicitly position themselves in opposition or as alternatives to mainstream media (Holt et al., 2019). These media differ significantly from legacy media in terms of ideological positioning, journalistic practices, and often in the dissemination of political content, including populist messaging, conspiracy narratives, or disinformation (Mayerhöffer, 2021; Mayerhöffer & Heft, 2022; Waisbord, 2022). As they often occupy critical spaces in public discourse, they thereby offer scholars crucial insights into broader societal and political dynamics on the fringes. As such, the alternative media observatory allows for the investigation of "alternative" or "fringe" agenda and opinion dynamics. As a digital platform designed to provide systematic, accessible, and comparable data about ANM across different national contexts, it is situated amongst computational explorations using data-driven methods to academic and professional fields associated with Social Sciences and Humanities (SSH) (Willaert et al., 2020, p. 2). As such, the observatory draws on previous scientific experiments with large corpora of text data (e.g., the Google Books corpus), as well as methods such as "distant reading" and "culturomics" aiming at extracting patterns from big textual data (Michel et al., 2011; Moretti, 2013).

The choice of the term "observatory" signals an infrastructure intended not merely as a passive data repository but rather as an active, interactive tool that allows researchers to observe, detect, and interpret patterns and changes in media content dynamically. The observatory does not resemble the more classic astronomical observatory that processes a constant, stable flow of data on natural objects (e.g., radiation) (Rogers, 2018). Unlike traditional databases or archives,

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the observatory invites continuous exploration and comparative interpretation, bridging qualitative, quantitative, and computational methodologies. Instead, the media observatory offers a way to capture "emergent" properties of societal and cultural phenomena on the fringes of society (Willaert et al., 2020).

As such, the approach steps away from a focus on "big data" which – and even more in the era of "large" language models – seems like a vague way to describe epistemological approaches to digital phenomena. Instead, the observatory bases itself on the notion of "distant data" thereby acknowledging the mediated character and scale of the data. Working with distant data is not about replicating close reading at scale – as implied by traditional distant reading approaches, which focus primarily on broad textual patterns – but rather about accepting abstraction as a productive methodological constraint. Unlike "big data", which emphasizes sheer data volume and computational capacity, distant data specifically foregrounds the mediated and constructed nature of data collection, deliberately embracing abstraction to enable meaningful comparison of alternative media landscapes across languages, outlets, and national contexts. The observatory thus offers a balanced middle-ground between fully automated computational analysis and qualitative research approaches. While qualitative analysis excels in detailed explorations of style, rhetoric, and meaning, its scalability can be limited (Chen et al., 2017). Computational approaches, on the other hand, excel in scalability and replicability but often at the cost of interpretive depth and contextual nuance (Grimmer & Stewart, 2013). The observatory conceptually seeks to integrate the interpretive strengths of qualitative methods with computational and quantitative scalability, supporting iterative exploration and hypothesis generation.

The "distance data" approach seeking to span qualitative and computational methods is not without challenges. As Törnberg and Uitermark (2021) show, many strands of computational

social science implicitly commit what Bhaskar (2013) calls the "epistemic fallacy," treating abstracted data models as if they were unfiltered windows onto social reality. Any model of a social system – in this case based on alternative media content – is necessarily a simplification – and, in critical-realist terms, an ontologically partial one that no accumulation of empirical detail can fully overcome (Hayles, 2010). To navigate this, the Alternative Media Observatory adopts a posture of critical methodological pluralism (Danermark et al., 2005): rather than pitting quantitative against qualitative, or computational against interpretive, it deliberately combines them considering their respective ontological assumptions (Zachariadis et al., 2013). Quantitative outputs – topic models, network metrics, or time-series trends – are never endpoints but should be seen as invitations to qualitative interrogation (Giglietto et al., 2012). Conversely, ethnographic readings and close textual analyses gain scale and comparative leverage when anchored in systematic data collection across outlets and languages.

This pluralist stance treats NLP pipelines, geospatial mappings, and in-depth interviews not as competing validation tools (triangulation) but as complementary ways of representing different phenomena. For instance, a topic model might surface a surge in climate-related framing on a Danish fringe site; a follow-up qualitative reading of sampled articles then uncovers the rhetorical strategies that automated labels alone would obscure.

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Commented [FMH10R9]: No this is sloppy from my side - it taps into the discussion of whether different methods is used to check each other or is used to complement each other as they can show different facets.

Triangulation - as I remember from qual teaching - is when you use different methods to cross-check findings, thereby increaing validity.
But please remove

By foregrounding the ontological limits of each method and weaving them into an integrated workflow, the observatory embraces a complexity-informed sensibility: it does not assume deterministic "clockwork lines" data patterns in the alternative media ecosystem but contingent, emergent patterns that invite ongoing exploration and hypothesis generation. In this way, the platform offers not just data – and not just stories – but a flexible, reflexive infrastructure for understanding alternative news media in all their messiness and dynamism.

As such, the aim is not to provide robust analytical endpoints, which is intangible without theory-driven progress, but descriptive and initial analytical starting points.

Methodological Considerations: Outlet Selection and Sampling

While the observatory approach is theoretically applicable to a broad range of alternative media types, the specific selection of outlets for any given implementation requires careful consideration of multiple factors.

First, the observatory must distinguish between the *types* of alternative media that are conceptually suitable for such analysis versus the *specific outlets* chosen for practical implementation. Building on operational definitions of alternative news media proposed by Holt and colleagues (2019) and Heft and colleagues (2020), our specific observatory is based on sampled digital-born alternative media that maintain relatively stable web infrastructures, publish content regularly, and operate with some degree of editorial consistency. These media define themselves in opposition to the perceived "societal" mainstream from either a traditional partisan (left-wing or right-wing) or a cross-partisan, anti-systemic positioning. These criteria exclude certain forms of alternative media such as ephemeral social media accounts, party-affiliated media, or offline-only publications that may be equally important to understand alternative media ecosystems but require different methodological approaches.

Commented [11]: in prolongation to first comment - I guess here it is clearly a decision taken by the specific observatory - while the first part might be seen as a requirement for all observatories?

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The selection process must also consider the sensitivity of the observatory approach to extreme outliers. Some alternative media outlets may publish at dramatically different frequencies or maintain distinct editorial profiles that could skew aggregate analyses. For instance, a single high-volume outlet might dominate temporal trends or topic distributions, potentially obscuring patterns from smaller but equally significant outlets. To mitigate this, the observatory design should include disaggregation capabilities, allowing users to filter by individual outlets or outlet categories, and to examine data both with and without potential

outliers. Additionally, the selection must account for potential biases in scraping accessibility. If the sampling procedures do not account for how web content is often *volatile* (i.e., subject to change), *personalized* (i.e., presented in response to request characteristics), and *unindexed* (i.e., not discoverable public search indices or rendered only via dynamic scripts), considerable sampling bias can be introduced (Foerderer, 2023). Certain types of alternative media may be more likely to implement technical barriers to scraping, such as sophisticated bot detection or dynamic content loading. This could introduce systematic biases where more technically sophisticated or well-resourced outlets are underrepresented in the dataset. Researchers must be transparent about these limitations and consider how they might affect the representativeness of their findings.

Many alternative media outlets blur the boundaries between news and opinion, making the distinction between different content types both methodologically and theoretically significant. A news-only scope yields cleaner categories but risks missing politically salient discourse embedded in commentary, opinion pieces or features. A more inclusive approach that captures all published content might better reflect the actual discursive practices of alternative media, while a more selective approach might provide cleaner analytical categories. The observatory should support disaggregation by content type, allowing researchers to examine news, opinion, and other content categories separately or in combination. This flexibility enables researchers to test hypotheses about how different types of content contribute to overall discursive patterns and to make informed decisions about which materials are most relevant to their specific research questions.

Commented [FMH13]: There was a comment here I deleted - I believe sampling do some extent both involves media and content, but we can discuss it if makes sense - this is a response to an earlier comment regarding why only "news content" was scrape, which I now admit seems odd or even stupid.

In sum, outlet and content sampling should be theory-driven, transparent, and bias-aware. Regarding media sampling, it is important to define inclusion/exclusion criteria, mitigate distortions by enabling disaggregation and sensitivity tests, audit scraping for volatility, personalization, and technical barriers, noting how these shape representativeness. In terms of

content, scope decisions (news-only vs. inclusive) are relevant and should be flexible in terms of using analytic filters rather than fixed constraints. Both steps require documentation of coverage, missingness, and technical limitations to justify how choices align with research aims.

The Nordic Pilot: Mapping ANM landscapes across the Nordic Countries

The Alternative Media Observatory specifically focuses on ANM landscapes in a cross-national context. As argued for in chapter 16 and 18, a key factor in cross-national comparative research relates to how national contexts shape journalism – especially in times of digitization and broader economic, societal and cultural shifts (Hanusch, 2019). An alternative journalistic landscape in a country is neither fixed, nor a homogenous entity (Maares & Hanusch, 2023). This idea is rooted in media systems theory concerned with the roles of institutionalized journalism in different national settings (Hallin & Mancini, 2004).

For this chapter, the ongoing work of a pilot Nordic observatory focusing on ANM content in Denmark, Finland, Norway, and Sweden will be used to explain the different steps of observatory building. As such, the Nordic observatory follows other studies on ANM content by comparing the output of different alternative media outlets in a cross-national setting (Heft et al., 2024; Kaiser et al., 2020; Mayerhöffer & Heft, 2022; McDowell-Naylor et al., 2023; Müller & Freudenthaler, 2022; Staender et al., 2024) rather than comparing ANM content against mainstream news media content. These countries share certain socio-political traits – relatively stable democratic structures and high-trust media landscapes – yet differ in media regulation, political culture, and historical trajectories of alternative media emergence. This selection promises to facilitate insightful comparative analyses, shedding light on how national contexts influence alternative media's discursive strategies, topical agendas, and ideological orientations.

Data Collection

Most ANM outlets do not put content behind paywalls, which is why their content is often manageable to scrape. However, a programming setup is still required to crawl URLs from main sites, ensure that the content is permissible to collect, as well as data cleaning and standardization. The *first* step involves creating a seed list with the main URL to each of the different outlets based on the sampling of ANM as discussed above. The Nordic observatory focuses exclusively on digital ANM with a clear anti-mainstream leaning (see Kristensen et al., 2023 for elaboration).

The *second* step consists in extracting links to all articles based on the folder structure on each news site. News outlets typically organize their web content using a hierarchical structure that includes subfolders corresponding to categories such as politics, culture, or economy (e.g., `/politics/`, `/opinion/`, `/world/`). These subfolders are often reflected in the site's sitemap – an XML file that lists all discoverable URLs on a website in a structured format. This step also involves ensuring that the scraping procedure is within the terms of service (ToS) of each outlet typically by checking the so-called `robots.txt`. This file is a simple text file hosted at the root of a domain that informs automated agents (e.g., search engine crawlers or research scrapers) about which parts of a site may or may not be accessed. Although not legally binding, these files represent a widely accepted standard for ethical data collection. More broadly, responsible digital scholarship requires balancing the scientific value of data collection with respect for platform autonomy, data minimization principles, and the avoidance of server overload. Researchers should document their scraping practices, ensure their data usage aligns with institutional ethical guidelines, and, where appropriate, seek permission from outlet administrators when uncertainty arises (Brown et al., 2024). In the *third* and final step, sub-page lists are cleaned to exclude non-article content.

Articles were scraped using `Trafilatura` (Barbaresi, 2021), a library optimized for large-scale text and metadata extraction. The scraper was implemented as a modular Python script with rate limiting, error handling, and rotating IPs to ensure stable and compliant collection. Unreachable pages were excluded after repeated failed attempts, and long-running jobs were managed via persistent server sessions. Trafilatura's built-in content extraction pipeline removes boilerplate, navigation elements, and other non-article artifacts so no extensive content cleaning is needed.

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Analytical Approach to ANM content

The corpus comprises 953,846 articles published between January 2008 to September 2025 from 67 different alternative news media outlets that all remain active in Denmark, Finland, Norway, and Sweden in 2025. To ensure data validity and quality, a systematic validation procedure was conducted. This involved randomly sampling a small sample of diverse articles (opinion pieces, news, advertisements) from each outlet and manually verifying the retrieved content and publication dates. To contextualize comparisons, each outlet was given a lightweight profile of its political orientation. This combined desktop research (e.g., self-descriptions in "about us" sections) with interim coding of a stratified sample of articles across key issues (e.g. migration, climate, gender). The result is a set of fuzzy, non-exclusive tags capturing recurring discursive stances, used as optional metadata.

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The first analytical step consists in applying descriptive statistical methods to establish baseline differences and structural features. Descriptive approaches – counts, temporal trends, and distributional summaries – are crucial for generating an initial overview and identifying outliers, gaps, and shifts over time. These patterns offer contextual grounding for more advanced computational methods such as network analysis or topic modelling, which benefit

from a clear sense of volume, frequency, and source concentration. This layered approach supports hypothesis formation, comparative insights and iterative exploration.

Table 1. Top 6 media outlets and total number of articles across the four countries (2005-25)

Denmark				Finland			
Domain	**Ideology**	**Count**	**Avg. year**	**Domain**	**Ideology**	**Count**	**Avg. year**
180grader	Right	124,247	6,539	Mvlehti	Right	59,599	4,966
24nyt	Right	55,316	6,146	Uvmedia	Right	10,548	1,318
Piopio	Left	40,680	2,404	Pt-media	Other	8,227	1,028
Denkorteavis	Right	28,482	2,034	Kansalainen	Right	7,954	530
Document	Right	22,282	2,025	Magneettimedia	Other	4,184	209
Arbejderen	Left	12,524	1,789	Verkkosanomat	Right	1,226	477
Norway				**Sweden**			
Domain	**Ideology**	**Count**	**Count**	**Domain**	**Ideology**	**Count**	**Avg. year**
Document	Right	136,786	6,406	Tidningensyre	Left	54,186	3,870
Inyheter	Right	28,647	5,729	Nyadagbladet	Other	35,233	2,516
Frihedskamp	Right	20,024	1,252	Bulletin	Right	34,450	5,741
Rights	Right	15,781	749	Samnytt	Right	30,136	3,348
Herland Report	Right	7,552	580	Nyheteridag	Right	17,763	1,366
Radikalpolitikk	Left	4,245	326	Nyatider	Right	17,207	1,229

Notes: Swedish outlet Swebbtv shows a high total count (34,778) and average count (4,640), but is excluded from the overview as it does not qualify as a text-based medium.

Table 1 shows the overall distribution of articles and reveals strong national variation in outlet productivity. Denmark, Finland and Norway show a "centralised" content production with a single outlet – 180grader.dk, mvlehti.net, and document.no – accounting for nearly half of the total articles per country. Sweden shows a more decentralised spread among top outlets like Tidningen Syre, Swebbtv, and Nyadagbladet, suggesting a more diversified ANM landscape. However descriptive, these findings indicate underlying structural differences worth investigating in a comparative perspective. These descriptive distributions make visible structural asymmetries (e.g. concentration vs diversification) that can inform comparative case selection or frame later qualitative inquiry into national media ecologies.

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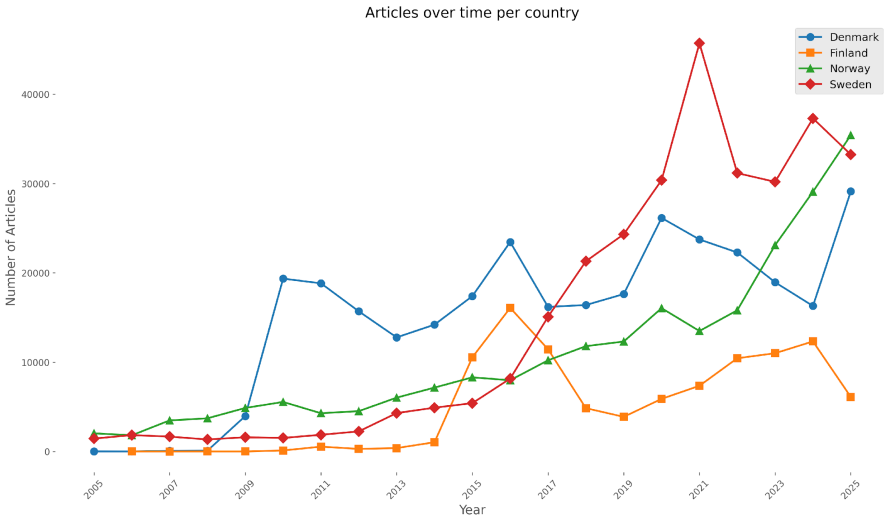
Figure 1. ANM Articles per country over time



Note: n= 953,846 articles from 67 different alternative news media outlets (2005-2025).

Figure 1 complements this picture with a longitudinal view. Denmark's ANM landscape surged around 2010 and remained relatively stable afterward, with minor fluctuations. Finland's growth trajectory is more gradual, gaining momentum after 2015 but maintaining consistently lower output. Norway shows a steady, gradual growth trajectory without notable peaks. Sweden shows sustained growth starting around 2015, peaking sharply in 2021 – a possible response to political or media-related events – before declining again in 2022. These patterns suggest distinct national dynamics in the development of alternative media ecosystems: Denmark's earlier consolidation, Sweden's more recent expansion, Norway's steady growth, and Finland's more modest growth. Taken together, such longitudinal patterns provide a comparative backdrop and invite questions about how political or regulatory shifts might relate to surges or declines in content, which can be probed further through e.g., content analysis (qualitative or quantitative) of periods where certain countries show patterns of interest.

For more advanced analysis, this chapter focuses specifically on domain co-occurrences in article content using network analysis, as illustrated in Figure 2 (see also chapter 13 in this volume), and overarching topical themes derived from topic modelling of article leads, depicted in Figure 3 (see also chapter 14 in this volume). These approaches demonstrate practical examples of leveraging ANM content within observatory construction. While network analysis of domains and topic modelling of content are highlighted here for illustrative purposes, the full Nordic Observatory includes additional analytical techniques and visualizations.

In analysing the interconnections between Nordic ANM outlets, a directed network based the sources used in the articles is constructed (Figure 2). The visualization reveals distinct communities (clusters) of scientific sources, mainstream news media, SoMe sources, as well as the position of the individual ANM with regards to these sources. It is worth highlighting how SoMe platfroms such as YouTube, Twitter/X (via t.co links), and Facebook is highly used in ANM content. This centrality underscores theoretical claims about the hybridization of alternative media landscapes, where traditional distinctions between mainstream, alternative, and social media increasingly blur (Mayerhöffer et al., 2024).

Figure 2. Network of Domain co-occurrences

Commented [20]: I am a bit unsure about the interpretation of the network graph - this is hyperlinks in articles, if I get it right - so really the type of analysis you are doing is much more in line with Heft et al. 2021 then with the curation piece - so I actually dont thinkt what it shows is the "prominent intermediary role of social media platforms" (unless there is something that I misunderstand here. It shows something about ANM sources

Commented [21]: since this is used for illustrative purposes, I think it is also totally fair that you dont have results. but what I think would be important for this chapter to give an indication of WHAT COULD BE DONE with this graph. first of all one would need to explain that what you now can do is investigating each colored cluster with focus on the type of hyperlinked sources these contain (mainstream, some, other ANM) that allows for typology building or a deep dive into certain clusters (like this, you also activate the quali-quant a bit more). If this doesnt make sense, lets chat :)

Commented [22]: isnt it like this really: the network analysis reveals
-types of ANM sources (the clusters): scientific sources, MSM, SoMe, and so on
- the position of individual ANM with regard to these source types?

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Commented [FMH24R23]: This is the curators of digital counterpublics-ref?

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Note: The network analysis was performed on hyperlinks extracted from article content. A directed graph was constructed where nodes represent domains and edges represent hyperlink references between domains. The network visualization uses a force-directed layout (Fruchterman-Reingold algorithm with k=0.3) to position nodes, with node sizes scaled by degree centrality. The analysis was conducted using NetworkX, and the network data was explored using Gephi.

Commented [26]: this is an interesting finding, but I think it is getting a bit too demanding here for people that dont know anything about network analysis - as said before, I think you should focus on outlining which results one can obtain from such an approach, rather than being very detailed about the actual results

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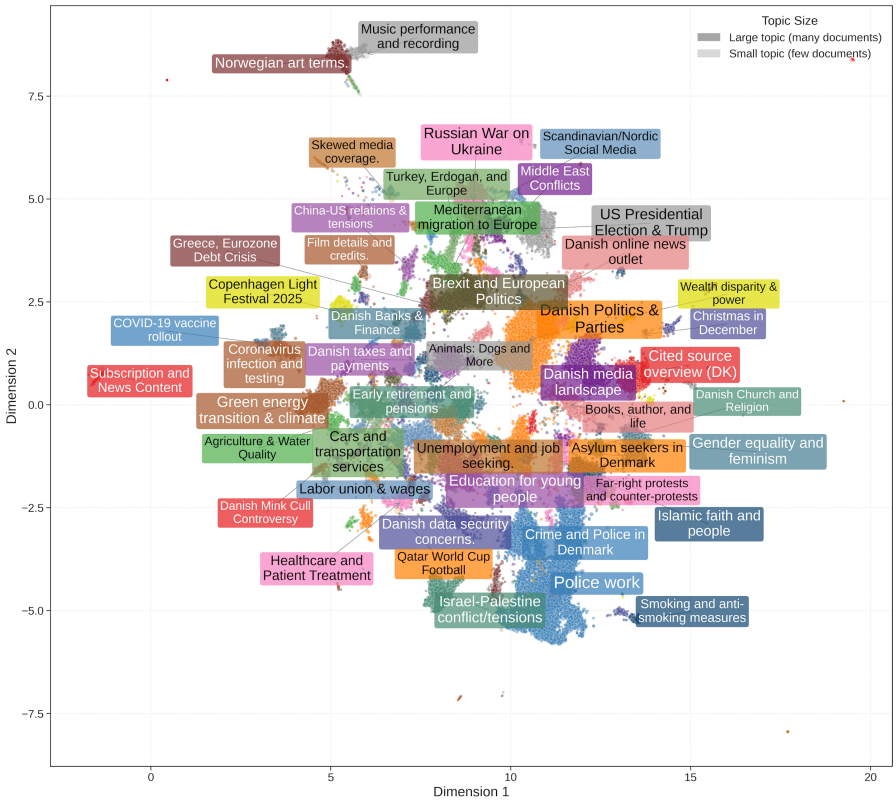
While Figure 2 serves primarily as an illustration, the clusters it reveals could form the basis for further inquiry. For example, one could investigate the composition of each cluster with attention to the mix of mainstream, social media, and other ANM sources, thereby building typologies of citation/source practices. Such an approach would also enable targeted qualitative deep-dives into specific clusters, activating the quali-quant potential of the observatory.

The last visualization (see Figure 3) is based on a topic model of content from Danish outlets (n=238,049) and represents a semantic map, where each point represents the article sub-headline and each labelled cluster represents a distinct topic. Using an embedding model trained only on Danish data (KennethTM/MiniLM-L6-danish-encoder-v2), the sub-headlines

are transformed into high-dimensional vectors that capture semantic meaning. These vectors are then projected into two-dimensional space using a dimensionality reduction algorithm (UMAP) for visualization purposes.

Figure 3. Topic Distribution in Embedding Space for Danish Alternative News Landscape 2010-2025



The visualization makes visible both large thematic clusters and more peripheral or unexpected groupings. These provide entry points for targeted close reading — for example, tracing how crime reporting is framed, or how international conflicts are connected to domestic debates. The size of each cluster corresponds to the volume of coverage, with larger topics containing more articles. This approach allows for the observation of broad thematic patterns and subtle

relationships between topics that might not be apparent through traditional content analysis methods. At the same time, results should be interpreted with caution (e.g., "Copenhagen Light Festival" appearing as a standalone cluster is quite misleading) and treated as an exploratory starting point for further qualitative analysis.

Ethical and Epistemological Considerations

The main ethical considerations of constructing a cross-national observatory of ANM content relate to the automated scraping of news articles. The observatory deals with public-yet-contentious discourse, often created by actors who either explicitly reject mainstream visibility or weaponize visibility to propagate their own agendas. As Brown and authors (2024) emphasize, scraping for research should not be seen as an ethically neutral technical practice, but rather as a methodologically embedded and normatively consequential practice that intersects with evolving legal, institutional, and epistemological concerns. Although all ANM sites in our dataset are publicly accessible and lack login walls or paywalls, their content is politically charged, and some outlets frame themselves as adversarial to media, academic, or governmental institutions, thereby fostering anti-systemic counterpublicity (Henriksen 2024).

Brown and authors (2024) argue that the key ethical challenge lies not merely in whether content is public, but in whether its collection and reuse respects contextual integrity – that is, the norms governing information flow within its original setting, e.g. Terms of Services (Nissenbaum, 2010; Zimmer, 2018). However, Brown and authors (2024) caution that even technically permissible scraping should not bypass broader ethical review. Here, a central consideration regards whether the subjects (editors) expect their actions (publishing) may be observed by researchers, which I believe can be justified given the increasing focus on ANM as a research field. One way to mitigate this is to submit the protocol to an Institutional Review Board (IRB). Yet, as noted in the AoIR guidelines (Franzke et al., 2020), IRB approval alone

is not sufficient for ethical assurance in digital trace research. Researchers must also reflect on how their own position, tools, and assumptions shape their knowledge production. For researchers engaging in scraping for the first time, colleagues can also be a resource for understanding and assessing the ethical dimensions in the particular environment, alongside navigating the specific actors who are involved in institutionalizing the considerations discussed above (Brown et al., 2024, p. 20).

Another ethical layer involves the politics of inclusion: which ANM outlets are indexed, which are excluded, and on what grounds. Drawing on existing conceptual definitions and previous research can be considered best practice here but filters the landscape through existing scholarly criteria that may obscure other relevant or emergent actors. Moreover, decisions about what counts as "alternative" are not neutral and influenced by normative perspectives (see chapters 2-4 in this volume). In privileging easily scrapable, text-based material and certain structural features (site stability, HTML accessibility), we exclude other sources of alternative information. As such, any scraping-based corpus reflects a "construct" shaped by technical, normative, and institutional filters rather than a "full" or "total" view of data.

Another point relates to how scraping inevitably entails data reduction: articles become tokens, topics, or graphs. While such abstractions support cross-national comparison, they risk flattening the rich discursive, rhetorical, and ideological complexity of ANM content. The observatory addresses this through a hybrid methodological posture: quantitative outputs (e.g., topic distributions) are presented as prompts for qualitative interrogation – not endpoints.

Commented [28]: yes - but precisely because of this, you need to reframe the analytical examples somewhat
1) tone down the result focus
2) focus on what type of (quantitative, mapping) insights the methods allows for and
3) how it opens up for qualitative investigation

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Conclusion

The Alternative Media Observatory offers an infrastructural and methodological advancement, responding directly to the predominance of single-country studies and the absence of scalable comparative frameworks within ANM research. By integrating computational, quantitative,

and qualitative methods into a unified platform, the Observatory does not merely map macro-level trends or document micro-level interpretive textures – it explicitly facilitates hypothesis generation and comparative analyses across diverse national contexts.

This chapter has demonstrated concretely how the Observatory operationalizes these ambitions through a pilot implementation in Denmark, Finland, Norway, and Sweden. The dataset is generated via transparent and replicable scraping of ANM articles and enables analysis through multilingual embeddings, clustering, and topic modeling, thus revealing discursive structures and cross-national patterns invisible in traditional content analyses. Importantly, these findings represent starting points rather than endpoints; the visualizations and data representations explicitly serve as prompts for hypothesis-driven comparative inquiry.

By embedding transparency, methodological documentation, and interactive exploration at its core, the Observatory speaks to broader epistemological calls for interpretability, accountability, and reproducibility in computational media research (Grimmer & Stewart, 2013; Törnberg & Uitermark, 2021). Moreover, the Observatory explicitly invites researchers to generate and test comparative hypotheses through its visualizations and interface, moving beyond merely descriptive mapping to actively shaping the research agenda within ANM studies. Ultimately, the Observatory strives to be more than a technical platform – it represents a conceptual shift toward reflexive, methodologically pluralist infrastructures capable of supporting sustained hypothesis-driven comparative research. By doing so, it provides a blueprint for more comprehensive, nuanced, and systematically comparative approaches to alternative news media research.

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